Data Mining - Final Report

Team 59

陳彥瑋

潘怡璇

賴琮運

蕭宜芳

Table of Contents

1. Instruction 4
    1.1 Competition Background 4
    1.2 Implementation we do 5
2. Related Works 6
    2.1 Ensemble Learning 6
        2.1.1 Stacked Generalization 6
        2.1.2 The Basic Principle of Stacked Generalization 6
    2.2 Feature Engineering 7
        2.2.1 Feature selection 7
        2.2.2 Special Challenges in Competitions 7
    2.3 Hyperparameter Tuning 8
3. Methodology 8
    3.1 Data Preprocessing 9
        3.1.1 Removal of Irrelevant Features 9
        3.1.2 Addition of Interaction Features 10
        3.1.3 Application of Autoencoders 11
        3.1.4 Missing Value Imputation 12
    3.2 Model Training and Optimization Process 13
        3.2.1 Data Preparation and Cross-Validation 13
        3.2.2 Model Training and Evaluation 13
        3.2.3 Kappa Optimization 13
        3.2.4 Final Prediction and Submission 14
    3.3 Multi-model Ensembling and Multiple Output 14
    3.4 Adjust hyperparameters 14
        3.3.1 Submission1 15
        3.3.2 Submission2 15
        3.3.3 Submission3 15
4. Experiments and Results 16
    4.1 Experiments 16
    4.2 Results 17
        4.2.1 Effectiveness of the Fusion Strategy 17
        4.2.2 Significance of the Evaluation Metric 17
        4.2.3 Competition Ranking Results 17
5. Discussions and Future Works 18
    5.1 Discussion 18
        5.1.1 Model Performance Analysis 18

5.1.2 Performance of Single Models 18
5.1.3 Effectiveness of the Ensemble Strategy 18
5.1.4 Optimization and Evaluation 19
5.1.5 Comprehensive Feature Engineering 19
5.1.6 Advanced Data Preprocessing 19
5.1.7 Custom Loss Function Optimization 19
5.2 Limitations 19
5.2.1 Data Constraints 19
5.2.2 Model Complexity 19
5.2.3 Evaluation Metric Challenges 20
5.3 Future Works 20
5.3.1 The issue of overfitting in deep learning models 20
5.3.2 The issue of class imbalance 20
5.3.3 The issue of model interpretability 20
6. Group Contribution 21
7. Reference 21

1. Instruction

1.1 Competition Background

With the rise of the digital age, the amount of time children and adolescents spend online continues to increase[1][2], bringing about a growing number of health and psychological issues. In particular, **Problematic Internet Use (PIU)**, or excessive reliance on the internet, has become a significant factor affecting the mental and physical health of children and adolescents[3]. Overuse of the internet can lead to problems such as anxiety, depression, decreased academic performance, and difficulties in interpersonal relationships.

Traditionally, identifying problematic internet use requires clinical assessments, which are time-consuming and require specialized tools[4][5]. Furthermore, the use of such tools may face barriers depending on cultural backgrounds and social conditions. Therefore, finding ways to detect problematic internet use early and intervene without the need for clinical expertise is an important issue to address in today's society.

This competition aims to develop a predictive model using children's physical activity data to identify early signs of problematic internet use. These physical activity data include accelerometer data (such as steps, acceleration, etc.) worn on the wrist, health measurements, and fitness test results, all of which are relatively easy to obtain and do not require clinical expertise. By analyzing these data, the goal is to predict the **Severity Impairment Index (SII)**, which indicates the degree of problematic internet use, and provide early intervention opportunities to promote healthier digital habits.

The goal of this competition is to predict the severity of problematic internet use in children and adolescents based on their physical activity data, helping to identify individuals at risk and providing an opportunity for early intervention.

1.2 Implementation we do

In this competition, we utilized multiple advanced models to make predictions, employing an ensemble learning strategy to enhance the overall accuracy of the model. We selected LightGBM, XGBoost, and CatBoost, which are highly effective for structured data, as well as TabNet, a deep learning model that automatically selects the most important features. To ensure stability, we employed cross-validation to evaluate the performance of each model and used QWK (quadratic weighted kappa) as the evaluation metric. During the prediction phase, we combined the predictions of multiple models using a weighted ensemble approach and adjusted the predictions to maximize accuracy. The integration of these strategies contributed to a better performance in the competition.

2. Related Works

2.1 Ensemble Learning

2.1.1 Stacked Generalization[8]

In the field of machine learning, improving the prediction accuracy of models has been a focal point of research. While traditional single learners may perform well in certain scenarios, their performance is often limited when faced with complex and dynamic data. As a result, researchers have proposed various ensemble learning methods designed to combine the predictions of multiple learners to enhance overall performance. Stacked Generalization, introduced by David H. Wolpert in 1992, is one of the most effective ensemble learning strategies, and it has since become an important tool for improving prediction accuracy.

2.1.2 The Basic Principle of Stacked Generalization

The core idea of Stacked Generalization is to use the outputs of multiple base learners (also referred to as "level 0 generalizers") as new features to train a higher-level learner (the "level 1 generalizer"). This way, the final prediction depends not only on the output of a single model but rather on the combined results of multiple learners. This process helps reduce errors and improves the accuracy of predictions. Unlike traditional ensemble methods, such as simple averaging or weighted averaging, Stacked Generalization uses the cross-validation of multiple base learners to estimate each learner's prediction error. This error information is then passed to the level 1 generalizer, which helps correct the errors. This enables Stacked Generalization to significantly improve prediction accuracy, especially when dealing with challenging and highly complex problems.

Stacked Generalization is a powerful ensemble learning method that holds great significance in the field of machine learning. Its core idea of multi-level learning and error correction enhances prediction accuracy by integrating the outputs of multiple learners. Stacked Generalization not only has deep theoretical value but also demonstrates excellent performance in practical applications, yielding significant results across various fields. As machine learning continues to advance, Stacked

Generalization is expected to play a key role in solving complex problems and improving prediction accuracy, and in our solution, we adopt it as our strategies.

2.2 Feature Engineering

In competitive machine learning and data science challenges, several key concepts frequently emerge, especially concerning feature selection[9], model evaluation, overfitting issues, and enhancing model stability. These challenges are prominent in fields such as bioinformatics, machine learning, and data mining. This literature review focuses on how feature selection techniques are applied in competitions, with a particular emphasis on the problems posed by high-dimensional data and small sample sizes, and how researchers have developed solutions for these issues.

2.2.1 Feature selection

Feature selection is a crucial preprocessing step[6][7], especially when dealing with high-dimensional data. The goal of feature selection is to reduce the dimensionality of the data, retaining the most informative features to improve model accuracy and reduce computational costs.

- **Filter Methods**: These methods assess the relevance of features using statistical properties, such as variance, correlation, or chi-square tests, independent of any learning algorithm. Filter methods are computationally efficient and widely used for high-dimensional datasets, such as microarray gene expression data or SNP datasets.
- **Wrapper Methods**: These methods evaluate feature subsets by training a model on them. For instance, genetic algorithms and ensemble techniques are employed to search through feature subsets to find the optimal set.
- **Embedded Methods**: Embedded methods combine feature selection with the model training process, such as L1 regularization in linear regression models and decision tree-based methods like Random Forests.

2.2.2 Special Challenges in Competitions

- **High-Dimensional and Small Sample Sizes**: In many fields, such as genomics, medical imaging, and text classification, the number of features

(dimensionality) far exceeds the number of samples. This presents a significant challenge and requires techniques such as feature selection, dimensionality reduction (like PCA), and regularization to handle the complexity effectively.

2.3 Hyperparameter Tuning

With the rapid development of machine learning (ML) technologies across various domains, how to effectively apply these techniques to solve real-world problems has become a key research focus. The performance of ML models largely depends on the choice of hyperparameters, which have a significant impact on model performance. However, in real-world applications, manually tuning hyperparameters requires considerable computational resources and time, especially as the volume of data increases. To reduce the burden of manual adjustment, Hyperparameter Optimization (HPO)[10] techniques have emerged and quickly become a core tool for improving the performance of machine learning models. This paper delves into the theoretical foundations and practical methods of HPO and discusses future research directions.

3. Methodology

For datasets with imbalanced data and a large amount of missing values, we proposed several methods to address these issues, aiming to improve the model's performance and prediction accuracy. During the process, we adopted the following steps to tackle these challenges.

Firstly, for feature selection, we utilized a correlation heatmap to identify variables that were less related to the target variable and eliminated them based on their correlation levels. This process not only effectively reduced the impact of irrelevant features on model performance but also shortened training time, making the model more concise and efficient. Subsequently, we introduced interaction features. By creating these interaction features, we were able to capture the underlying complex relationships between features, thereby enhancing the model's predictive capability. This allowed us to retain the features that had the most significant impact on the prediction results. Additionally, we addressed the challenge of high-dimensional data processing using an autoencoder, which mapped the high-

dimensional input data into a low-dimensional latent space, achieving data compression and feature extraction. The advantage of this method lies in its ability to learn the intrinsic structure and features of the data without requiring labeled data. For handling missing values, we adopted the KNN algorithm, which predicts missing values based on the characteristics of similar samples in the data. This approach effectively reflects the data structure and significantly improves the accuracy of imputation.

During the model training process, despite performing missing value imputation, the presence of a large amount of missing data still posed a risk of overfitting, which could reduce the model's generalization ability on unseen data. To address this issue, we adopted an ensemble learning approach and conducted multiple evaluations of model outputs. This strategy effectively mitigates the bias that may arise from individual models, thereby improving the overall prediction accuracy and stability. The process of multiple evaluations also allows for fine-tuning and optimizing the predictive capabilities of the model.

Below, I will provide a detailed explanation of our model's workflow, outlining the specific operations and results of each step. The goal is to present concrete and effective solutions for addressing the challenges of data imbalance and missing values.

3.1 Data Preprocessing

3.1.1 Removal of Irrelevant Features

Using a correlation heatmap, we identified features with low correlation to the target variable and removed them. This process helps simplify the model structure, reduces unnecessary computational overhead, and improves the efficiency of training and prediction. Eliminating irrelevant features also effectively lowers the risk of overfitting, enhancing the model's generalization ability on unseen data. Additionally, removing unrelated features improves the interpretability of the model, allowing us to better understand the role of each feature while accelerating the training proces

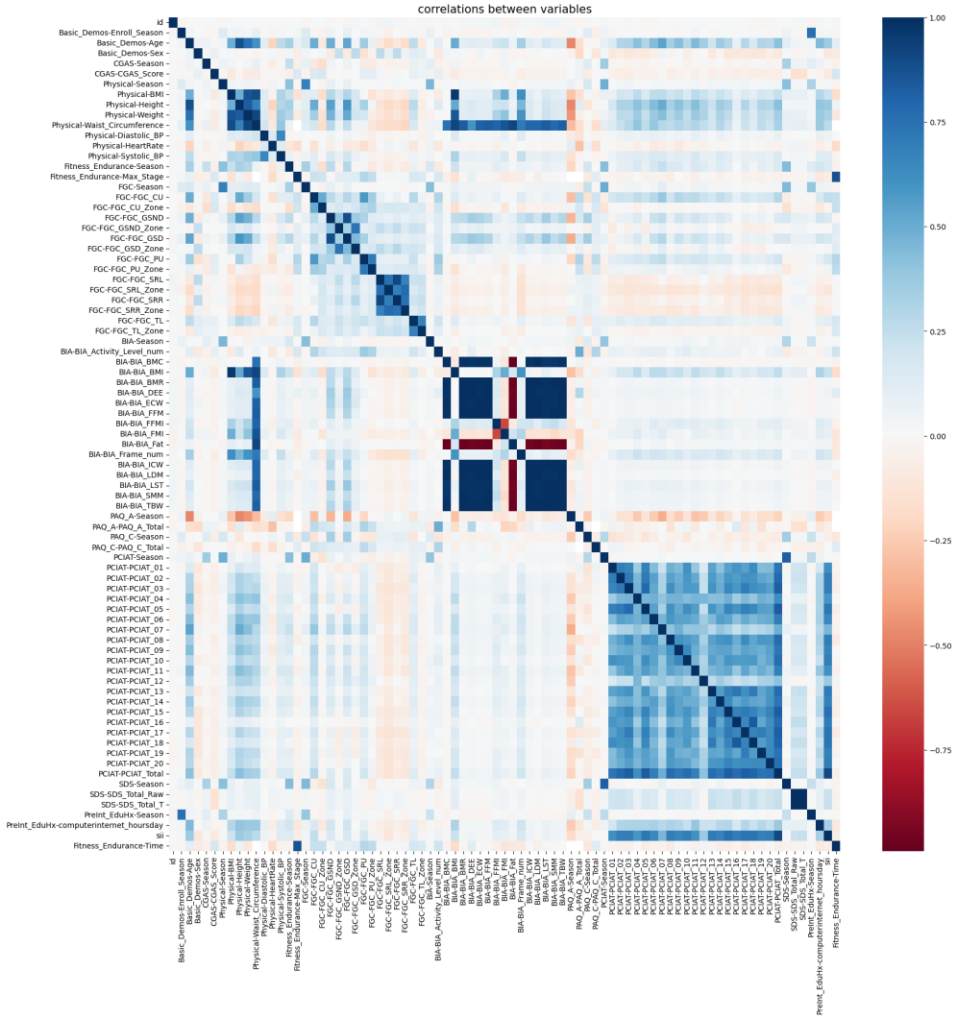


Figure 1. correlation heatmap

3.1.2 Addition of Interaction Features

Some features may exhibit strong interactions that cannot be fully captured by the model when using the original features alone. This is because models typically learn the individual effects of each feature, while in the real world, interactions between features often provide valuable additional information. Adding interaction features helps the model capture these complex relationships, thereby improving predictive performance.

For instance, interaction features like BMI_Age (the product of BMI and age) and Internet_Hours_Age (the product of daily internet hours and age) can better reflect the relationships between two features. The relationship between BMI and age, for example, may not be merely additive; their combination might reveal deeper impacts on health. Similarly, the interaction between age and internet usage may

uncover differing behavioral patterns across age groups, which are crucial for the model's predictive accuracy.

By introducing these interaction features, the model can more comprehensively capture the complex relationships between variables, reducing reliance on individual feature effects. This approach enhances the model's predictive accuracy and interpretability. Moreover, these interaction features not only improve model performance but also provide insights into the underlying associations among variables, aiding in a deeper understanding of the data.

```
def feature_engineering(df):
    season_cols = [col for col in df.columns if 'Season' in col]
    df = df.drop(season_cols, axis=1)
    df['BMI_Age'] = df['Physical-BMI'] * df['Basic_Demos-Age']
    df['Internet_Hours_Age'] = df['PreInt_EduHx-computerinternet_hoursday'] * df['Basic_Demos-Ag
e']
    df['BMI_Internet_Hours'] = df['Physical-BMI'] * df['PreInt_EduHx-computerinternet_hoursday']
    df['BFP_BMI'] = df['BIA-BIA_Fat'] / df['BIA-BIA_BMI']
    df['FFMI_BFP'] = df['BIA-BIA_FFMI'] / df['BIA-BIA_Fat']
    df['FMI_BFP'] = df['BIA-BIA_FMI'] / df['BIA-BIA_Fat']
    df['LST_TBW'] = df['BIA-BIA_LST'] / df['BIA-BIA_TBW']
    df['BFP_BMR'] = df['BIA-BIA_Fat'] * df['BIA-BIA_BMR']
    df['BFP_DEE'] = df['BIA-BIA_Fat'] * df['BIA-BIA_DEE']
    df['BMR_Weight'] = df['BIA-BIA_BMR'] / df['Physical-Weight']
    df['DEE_Weight'] = df['BIA-BIA_DEE'] / df['Physical-Weight']
    df['SMM_Height'] = df['BIA-BIA_SMM'] / df['Physical-Height']
    df['Muscle_to_Fat'] = df['BIA-BIA_SMM'] / df['BIA-BIA_FMI']
    df['Hydration_Status'] = df['BIA-BIA_TBW'] / df['Physical-Weight']
    df['ICW_TBW'] = df['BIA-BIA_ICW'] / df['BIA-BIA_TBW']

    return df
```

Figure 2. Feature Integration

3.1.3 Application of Autoencoders

To address the challenge of high-dimensional data, we propose a data dimensionality reduction and feature learning method based on autoencoders. As an unsupervised learning model, autoencoders are capable of mapping high-dimensional input data into a lower-dimensional latent space, achieving both data compression and feature extraction. The advantage of this approach lies in its ability to function without labeled data, while automatically learning the inherent structure and features of the data.

3.1.3.1 Data Preprocessing and Model Construction

First, the raw data is standardized to eliminate scale differences between features.

Subsequently, an autoencoder model is constructed, consisting of an encoder and a decoder. Through learning the data compression and reconstruction process, the model generates a lower-dimensional data representation. During training, mean squared error (MSE) is used as the loss function, and the Adam optimizer is selected for parameter updates.

3.1.3.2 Training and Encoding Generation

After training, the trained encoder is used to infer the data and generate low-dimensional encoded representations. These encoded data can be used as input for subsequent analysis or other machine learning models.

```
def perform_autoencoder(df, encoding_dim=50, epochs=50, batch_size=32):
    scaler = StandardScaler()
    df_scaled = scaler.fit_transform(df)

    data_tensor = torch.FloatTensor(df_scaled)

    input_dim = data_tensor.shape[1]
    autoencoder = AutoEncoder(input_dim, encoding_dim)

    criterion = nn.MSELoss()
    optimizer = optim.Adam(autoencoder.parameters())

    for epoch in range(epochs):
        for i in range(0, len(data_tensor), batch_size):
            batch = data_tensor[i : i + batch_size]
            optimizer.zero_grad()
            reconstructed = autoencoder(batch)
            loss = criterion(reconstructed, batch)
            loss.backward()
            optimizer.step()

        if (epoch + 1) % 10 == 0:
            print(f'Epoch [{epoch + 1}/{epochs}], Loss: {loss.item():.4f}]')

    with torch.no_grad():
        encoded_data = autoencoder.encoder(data_tensor).numpy()

    df_encoded = pd.DataFrame(encoded_data, columns=[f'Enc_{i + 1}' for i in range(encoded_data.s
hape[1])])

    return df_encoded
```

Figure 3. Encoding Generator

3.1.4 Missing Value Imputation

We use the KNNImputer method to fill missing values, which leverages the K-nearest neighbors algorithm to impute missing data based on the similarity between samples. We set the KNNImputer to impute missing values by considering the 5 most

similar neighboring samples (n_neighbors=5) and using their known values to predict the missing ones. This approach improves the accuracy of the imputation while preserving the overall structure and features of the data.

3.2 Model Training and Optimization Process

We employed cross-validation for model training and optimization to improve the predictive accuracy of the model in multi-class tasks, particularly in the case of class imbalance. The specific process is as follows:

3.2.1 Data Preparation and Cross-Validation

First, the features and target variable are extracted from the training set train, where X represents the feature matrix and y represents the target variable, sii. Then, Stratified K-Fold cross-validation is applied to divide the dataset into n_splits folds, ensuring that each fold maintains the same distribution of the target variable. The training and validation sets in each fold are used to train and test the model, respectively.

3.2.2 Model Training and Evaluation

In this step, we train the model using the training set from each fold of the cross-validation process. For each fold, the model is trained on the training subset and evaluated on the validation subset. The model's performance is assessed using specific metrics, such as Quadratic Weighted Kappa (QWK), which is used to evaluate the accuracy of the model's predictions in multi-class classification tasks. The QWK score is calculated for both the training set and validation set after each fold. This process helps identify the model's strengths and weaknesses, and ensures that the model generalizes well across different subsets of the data.

3.2.3 Kappa Optimization

After completing cross-validation, we further optimize the Kappa score to enhance model performance. To achieve this, we apply a numerical optimization method (Nelder-Mead) to adjust the prediction results and select the best tuning

parameters to maximize the final Kappa score. The optimization process is performed using the non-rounded predictions (oof_non_rounded), and generates the optimal prediction results. The core of this process is fine-tuning the model's predictions to minimize prediction errors and improve overall prediction accuracy.

3.2.4 Final Prediction and Submission

The adjusted predictions are rounded to the nearest integer, generating the final predicted labels. The final predictions, along with the sample IDs, are compiled into a submission file, ready to be submitted to the competition platform. The advantage of this method lies in the combination of cross-validation, Kappa evaluation, and optimization steps, which effectively enhance the model's performance in multi-class classification tasks, especially when dealing with class imbalance or ambiguous boundaries.

3.3 Multi-model Ensembling and Multiple Output

We employed ensemble learning and multiple output evaluations to enhance the stability and accuracy of our predictions. The core idea behind ensemble learning is to combine the predictions of multiple models using weighted averaging or other fusion strategies, reducing the bias and randomness inherent in individual models, thereby improving the final prediction results.

3.4 Adjust hyperparameters

We chose Optuna to find the best hyperparameters because it offers an efficient and flexible approach for hyperparameter optimization. Optuna automates the search process using advanced sampling techniques like TPE (Tree-structured Parzen Estimator), which allows it to explore a wide parameter space and converge on the best parameters quickly. Its ability to handle complex search spaces, use pruning to discard non-promising trials early, and support parallelization makes it particularly suitable for optimizing models efficiently. This helps improve the model's performance while saving computational resources. Therefore, through this method, we performed a search for each ensemble model to identify the optimal hyperparameters for each model individually.

3.3.1 Submission1

Firstly, in the first stage (Submission1), we selected six models: LightGBM, XGBoost, CatBoost, TabNet, Random Forest, and Gradient Boosting, and assigned different weights to each model (4.0, 4.0, 5.0, 4.0, 3.0, 3.0, respectively). These weight allocations were based on the performance of each model on the validation set, ensuring that the final prediction result accurately reflects the overall capability of the models. As a result, this ensemble model achieved a Quadratic Weighted Kappa (QWK) score of 0.53, demonstrating the best performance.

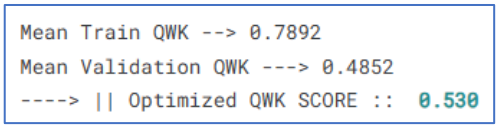


Figure 4. Submission1 Evaluation

3.3.2 Submission2

Secondly, in the second stage (Submission2), we selected only three models—LightGBM, XGBoost, and CatBoost—for the ensemble. The resulting QWK score was 0.457. In this stage, the number of models was reduced, but the models had lower correlation, aiming to explore whether a more streamlined model structure could achieve similar results.

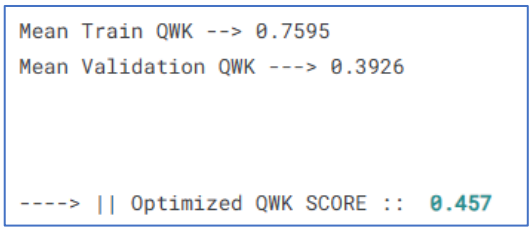


Figure 5. Submission2 Evaluation

3.3.3 Submission3

Thirdlly, in the third stage (Submission3), we expanded the selection of models and used seven regression models, including LGBMRegressor, XGBRegressor, CatBoostRegressor, RandomForestRegressor, GradientBoostingRegressor, ExtraTreesRegressor, and AdaBoostRegressor, for further ensemble. However, the QWK score for this approach was 0.452, indicating that the addition of more models did not necessarily lead to an improvement in performance.

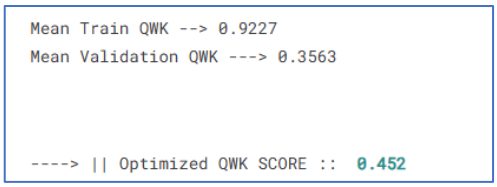


Figure 6. Submission3 Evaluation

Finally, we consolidated the prediction results from all three ensemble models into a single CSV file and applied a majority voting method to select the most frequent prediction as the final result. This multi-output strategy effectively reduces the random errors from individual ensemble models, further enhancing the stability of the predictions. Overall, the ensemble and multi-output approach demonstrated its robustness in scenarios involving multiple model fusion, especially in environments with class imbalance or significant data variations.

4. Experiments and Results

4.1 Experiments

In our experiments, we utilized five-fold cross-validation to evaluate model performance and used Quadratic Weighted Kappa (QWK) as the core evaluation metric to measure the consistency between predictions and actual values. The baseline tests of individual models indicated that CatBoost and XGBoost performed best on the dataset, attributable to their ability to handle non-linear data and categorical features effectively. However, individual models struggled to fully capture the complex relationships between multidimensional features and the target variable, resulting in certain limitations in their predictive capabilities. To enhance predictive accuracy and stability, we conducted multiple rounds of model fusion. In the first round, we combined LightGBM, XGBoost, and CatBoost using a Voting Regressor to aggregate model outputs, significantly improving overall performance. Subsequently, in the second round, we introduced Random Forest to the ensemble, further increasing the diversity of the fused model and ultimately creating a more robust ensemble.

4.2 Results

The final test results demonstrated that the fusion strategy effectively reduced bias, improved consistency, and delivered stable predictive performance. The final QWK evaluation result was 0.492, demonstrating the superiority of the fused model in addressing the prediction of problematic internet use scores. While there remains room for improvement in achieving ideal prediction consistency, this score validates the effectiveness of our approach.

4.2.1 Effectiveness of the Fusion Strategy

The fusion strategy demonstrated significant advantages in reducing bias and improving overall performance. By leveraging the strengths of different models, we effectively enhanced the adaptability of the model to the dataset and its ability to capture the relationships with the target variable. The results showed that the fused model outperformed individual models in terms of both accuracy and robustness, underscoring the applicability of diverse model combinations in addressing such problems.

4.2.2 Significance of the Evaluation Metric

As the core evaluation standard, QWK effectively demonstrated the consistency between the model's predictions and the actual values. Although the final score of 0.492 does not represent perfect consistency, it is significantly better than random agreement, indicating that the methods employed in this study can effectively handle such multidimensional datasets and show potential for further optimization.

4.2.3 Competition Ranking Results

Finally, our ranking was 714th. As shown in the figure below.

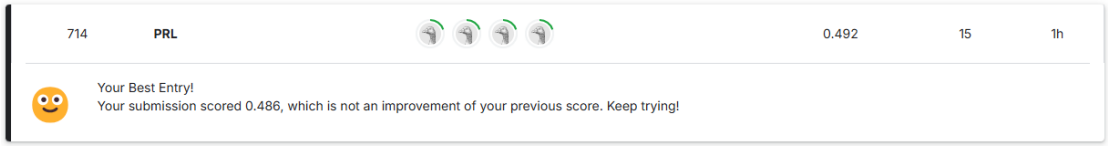


Figure 7. Competition Ranking Results

5. Discussions and Future Works

5.1 Discussion

The results of this study demonstrate the potential of our approach in predicting Problematic Internet Use Scores (SII). This section discusses model performance, methodological strengths, and the limitations of the research.

5.1.1 Model Performance Analysis

Our integrated approach, combining multiple models (LightGBM, XGBoost, CatBoost, Random Forest, Gradient Boosting, and TabNet), showed promising results in enhancing predictive accuracy and stability. Key observations include:

5.1.2 Performance of Single Models

CatBoost and XGBoost performed the best on the dataset, primarily due to their ability to handle non-linear data and categorical features. However, single models were limited in capturing the complex relationships between multi-dimensional features and the target variable.

5.1.3 Effectiveness of the Ensemble Strategy

The multi-stage ensemble strategy effectively leveraged the strengths of different models. In the first stage, the integration of LightGBM, XGBoost, and CatBoost through a Voting Regressor significantly improved overall performance. The second stage introduced Random Forest and Gradient Boosting to further increase model diversity, resulting in a more robust ensemble model. Final results demonstrated that the ensemble approach effectively reduced bias and improved consistency and stability.

5.1.4 Optimization and Evaluation

We used the Quadratic Weighted Kappa (QWK) as the primary evaluation metric to quantify the consistency between predicted results and true values. While the final QWK score leaves room for improvement, it validates the effectiveness of our method, achieving performance well above random consistency levels.

5.1.5 Comprehensive Feature Engineering

Creating derived variables, such as the interaction between BMI and age, and behavior-related ratios, helped capture subtle relationships in the data.

5.1.6 Advanced Data Preprocessing

Employing KNN imputation for missing values, feature standardization, and categorical encoding ensured data quality and consistency.

5.1.7 Custom Loss Function Optimization

Threshold tuning based on QWK improved the classification performance by better aligning continuous predictions with ordinal outputs.
Study Limitations

5.2 Limitations

5.2.1 Data Constraints

While the dataset offered diversity, its imbalanced class distribution posed challenges for model learning. Although SMOTE was applied for data augmentation, class imbalance in high-dimensional spaces remains a potential issue.

5.2.2 Model Complexity

While the ensemble strategy improved performance, it increased computational costs and training time, which might challenge real-time applications or large-scale

deployment.

5.2.3 Evaluation Metric Challenges

Although QWK is a robust evaluation metric, achieving ideal scores remains challenging in cases of overlapping class boundaries or noisy data.

5.3 Future Works

5.3.1 The issue of overfitting in deep learning models

This study has demonstrated significant effectiveness in the fusion of machine learning models while also revealing the limitations of deep learning models, particularly the issue of overfitting.

Future research could address this challenge by exploring more effective regularization techniques, such as Dropout or early stopping, or by adopting more lightweight deep learning architectures to balance model performance and stability. Considering the risk of overfitting, future efforts could also focus on enhancing the capabilities of traditional machine learning models, such as refining hyperparameter tuning methods for gradient-boosted models to further improve accuracy.

5.3.2 The issue of class imbalance

The issue of class imbalance remains a key area for future exploration. While techniques like SMOTE have shown promising results, further investigation into methods based on Generative Adversarial Networks (GANs) could generate more precise synthetic data, enhancing the model's predictive power for minority classes. Additionally, researching novel loss functions, such as weighted loss or focal loss, could contribute to improved model performance.

5.3.3 The issue of model interpretability

Improving model interpretability will also be a priority in subsequent research. Integrating model explanation tools such as SHAP or LIME could provide clearer insights into the relationships between features and predictions, which is crucial for practical applications in behavior prediction. For handling time-series data, future studies could consider leveraging more efficient sequence modeling techniques, such

as hybrid models combining traditional approaches with simplified architectures based on attention mechanisms, to capture both long-term and short-term patterns in sequential data.

In summary, addressing overfitting, enhancing class balance strategies, and improving model interpretability will form the foundation for advancing predictive accuracy and robustness in future studies.

6. Group Contribution

Members	Work	Contribution
陳彥瑋	EDA, Compile the written report and related work section	25%
潘怡璇	Build models, Methodology section	25%
賴淙運	data preprocessing, oral presentation PPT	25%
蕭宜芳	tune hyperparameters, result section	25%

7. Reference

[1] Andreassen, C. S., Pallesen, S., & Griffiths, M. D. (2017). The relationship between addictive use of social media, narcissism, and selfesteem: Findings from a large national survey. Addictive Behaviors, 64, 287–293.

[2] Kemp, S. (2019). Digital in, 2019 Retrieved July 01, 2019

[3] Daskalaki, E., Psaroudaki, K., Karkanaki, M., & Fragopoulou, P. (2020). Understanding the online behavior and risks of children: results of a large-scale national survey on 10-18 year olds. *arXiv preprint arXiv:2008.10274*.

[4] Savci, M., Tekin, A., & Elhai, J. D. (2020). Prediction of problematic social media use (PSU) using machine learning approaches. *Current Psychology*, *41*(5), 2755–2764.

[5] Srividya, M., Mohanavalli, S., & Bhalaji, N. (2018). Behavioral Modeling for Mental Health using Machine Learning Algorithms. *Journal of Medical Systems*, *42*(5).

[6] Stańczyk, U. (2014). Feature evaluation by filter, wrapper, and embedded approaches. In *Studies in computational intelligence* (pp. 29–44).

[7] Jović, A., Brkić, K., & Bogunović, N. (2015, May). A review of feature selection methods with applications. In *2015 38th international convention on information and communication technology, electronics and microelectronics (MIPRO)* (pp. 1200-1205). Ieee.

[8] Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, *5*(2), 241–259.

[9] Yvan Saeys, Iñaki Inza, Pedro Larrañaga,(2007). A review of feature selection techniques in bioinformatics, *Bioinformatics*, Volume 23, Issue 19,

[10] Yang, L., & Shami, A. (2020). On hyperparameter optimization of machine learning algorithms: Theory and practice. *Neurocomputing*, *415*, 295–316.